

Review Question Example #1

Q: Suppose during training you observe that the training loss oscillates wildly and sometimes increases dramatically between steps.

Think about:

- ▶ What is the most likely cause?
- ▶ What adjustment would you make?

Take 2–3 minutes to think before we discuss.

Example#1 | **Bad Answer** $\approx 0.5/3$ credit

Q: Suppose during training you observe that the training loss oscillates wildly and sometimes increases dramatically between steps. What is the most likely cause? What adjustment would you make?

"A: The model is unstable. I would try changing some hyperparameters."

- ▶ Why this is a **bad answer**: too vague
 - ⊙ **The problem:** Does not show understanding of the underlying cause (e.g., large learning rate leading to divergence)
 - ⊙ **The solution:** No proposal of a mechanism we learned about in class (e.g., learning rate adjustment)

Example #1 | **Partial Credit** $\approx 1.5/3$ credit

Q: Suppose during training you observe that the training loss oscillates wildly and sometimes increases dramatically between steps. What is the most likely cause? What adjustment would you make?

"A: The learning rate is probably too high. I would reduce the learning rate."

► Why this earns **partial credit**:

- ⊙ **The problem:** Correctly identifies a plausible cause (large learning rate)
- ⊙ **The solution:** Proposes a reasonable fix, but with **insufficient explanation** of why it would work (e.g., how reducing the learning rate would stabilize training)

Example #1 | Full Credit 3 / 3

Q: Suppose during training you observe that the training loss oscillates wildly and sometimes increases dramatically between steps. What is the most likely cause? What adjustment would you make?

"A: The learning rate is likely too large. With a large step size, the gradient updates overshoot the minimum, causing the loss to increase or oscillate between steps. Reducing the learning rate would help stabilize training by allowing smaller updates that are less likely to diverge."

► Why this earns **full credit**:

- ⦿ **The problem:** Identifies the correct cause
- ⦿ **The mechanism:** Explains **why** oscillation occurs (overshooting the minimum)
- ⦿ **The solution:** Proposes an approach (reduce learning rate) and explains **why** it would help (stabilize training by smaller updates)

Review Question Example #2

Q: Suppose you use **beam search** for text generation. In what type of task might this be **not preferred** compared to sampling-based methods? Give an example task and explain why.

Think about:

- ▶ What type of task?
- ▶ Why would beam search perform poorly?

Take 2–3 minutes to think before we discuss.

Example #2 | **Bad Answer** $\approx 1/4$ credit

Q: Suppose you use beam search for text generation. In what type of task might this be not preferred compared to sampling-based methods? Give an example task and explain why.

"A: Beam search is bad for creative tasks."

- ▶ Why this is a **bad answer**: too vague
 - ⊙ **The task**: Does **not clearly distinguish open-ended vs constrained tasks as we discussed in class**
 - ⊙ **Missing reasoning about mechanism (how beam search works)**: No explanation of how beam search behaves (e.g., tends to produce high-likelihood but generic outputs) and why that would be a problem for open-ended tasks (e.g., creative writing, story generation)

Example #2 | **Partial Credit** $\approx 3/4$ credit

Q: Suppose you use beam search for text generation. In what type of task might this be not preferred compared to sampling-based methods? Give an example task and explain why.

“A: In open-ended tasks like story generation, beam search can produce repetitive or generic outputs because it keeps the highest-probability sequences.”

- ▶ Why this earns **partial credit**:
 - ◉ **The task:** Correctly identifies open-ended generation
 - ◉ **The mechanism:** notes that beam search keeps high-probability sequences, which can lead to generic outputs
 - ◉ **Missing:** Does not contrast with translation/summarization

Example #2 | Full Credit 4 / 4

Q: Suppose you use beam search for text generation. In what type of task might this be not preferred compared to sampling-based methods? Give an example task and explain why.

"Beam search may not be preferred in open-ended tasks such as story writing or dialogue generation. Because it maximizes sequence-level likelihood, it often produces generic or repetitive high-probability outputs. In open-ended tasks, many continuations are valid, and sampling allows greater diversity. In contrast, for constrained tasks like machine translation or summarization, where correct outputs are limited, beam search is more appropriate since likelihood better aligns with correctness."

Example #2 | why **Full Credit** 4 / 4

- ▶ Why it earns **full credit**:
 - ◉ **Task distinction:** Open-ended vs constrained tasks
 - ◉ **Mechanism:** Explains sequence-level likelihood maximization
 - ◉ **Reasoning:** Connects likelihood bias to generic outputs
 - ◉ **Contrast:** Explains why beam search works well in translation/summarization

Review Question Example #3

Q: Why can applying normalization before a residual connection (Pre-Norm) stabilize training of deep transformers?

Think about:

- ▶ What problem does Pre-Norm address?
- ▶ What is the mechanism by which it helps?

Take 2–3 minutes to think before we discuss.

Example #3 | **Bad Answer** $\approx 0.5/3$ credit

Q: Why can applying normalization before a residual connection (Pre-Norm) stabilize training of deep transformers?

“Pre-Norm makes training more stable and improves performance.”

- ▶ Why this is a **bad answer**: too vague
 - ◉ **The problem**: Does not identify vanishing gradients
 - ◉ **The mechanism**: No explanation of gradient flow
 - ◉ **The reasoning**: Simply restates the question

Example #3 | **Partial Credit** $\approx 1.5/3$ credit

Q: Why can applying normalization before a residual connection (Pre-Norm) stabilize training of deep transformers?

“Pre-Norm helps prevent vanishing gradients by normalizing activations before they go into the layer, which makes training more stable.”

- ▶ Why this earns **partial credit**:
 - ⊙ **The problem:** Mentions vanishing gradients
 - ⊙ **The mechanism:** Notes normalization applied to activations before the layer
 - ⊙ **Missing:** Does not explain the role of the residual connection and how the identity path aids gradient flow

Example #3 | Full Credit 3 / 3

Q: Why can applying normalization before a residual connection (Pre-Norm) stabilize training of deep transformers?

"In very deep transformers, gradients can vanish or explode as they pass through many nonlinear layers. In Pre-Norm, normalization is applied before the transformation $F(\cdot)$, so the block becomes $y = x + F(\text{LN}(x))$ instead of post-Norm $y = \text{LN}(x + F(x))$.

This keeps the residual (identity) path x unchanged, there is always a direct path for gradients to flow back through the identity connection, which helps prevent vanishing gradients even if $F(\cdot)$ causes gradients to vanish.

Example #3 | why **Full Credit** 3 / 3

- ▶ Why this earns **full credit**:
 - ◉ **The problem:** Identifies vanishing/exploding gradients
 - ◉ **The mechanism:** Explains normalization placement
 - ◉ **Residual insight:** Highlights preserved identity path (how layer norm interacts with residual connection)

Review Question Example #4

Q: In post-training, why does maximizing likelihood with supervised fine-tuning (SFT) not necessarily maximize human preference?

Think about:

- ▶ What does likelihood optimize?
- ▶ Why might that differ from what humans prefer?

Take 2–3 minutes to think before we discuss.

Example #4 | **Bad Answer** $\approx 0.5/3$ credit

Q: In post-training, why does maximizing likelihood with SFT not necessarily maximize human preference?

“Because likelihood is different from preference.”

- ▶ Why this is a **bad answer**: circular and vague
 - ⊙ Does not explain what likelihood optimizes
 - ⊙ Does not explain what human preference depends on

Example #4 | **Partial Credit** $\approx 1.5/3$ credit

Q: In post-training, why does maximizing likelihood with SFT not necessarily maximize human preference?

“SFT trains the model to imitate labeled data by maximizing the probability of the reference responses. But human preference may depend on whether responses are helpful or safe.”

- ▶ Why this earns **partial credit**:
 - ◉ Correctly states SFT imitates demonstrations
 - ◉ Mentions preference may involve helpfulness or safety
 - ◉ **Missing:** Does not explain token-level objective
 - ◉ **Missing:** Does not explain ambiguity or multiple valid responses

Example #4 | Full Credit 3 / 3

Q: In post-training, why does maximizing likelihood with SFT not necessarily maximize human preference?

“Supervised fine-tuning maximizes token-level likelihood of the reference responses in the dataset. This objective rewards matching the exact tokens in demonstrations.

However, many prompts have multiple valid responses, and human preference depends on sequence-level qualities such as helpfulness, harmlessness, and honesty. As a result, a response can have high likelihood under the training data but still be less preferred by humans for being unhelpful, unsafe, or dishonest (e.g., hallucinated)

- ▶ Why this earns **full credit**:
 - ⊙ Explains token-level likelihood objective
 - ⊙ Identifies sequence-level human preference criteria
 - ⊙ Clearly states objective mismatch